

## Task 1:

Idea generation:

What do I like? What am I interested in?

Games

Art

Cars

Swordsmanship

Martial Arts (Specifically kickboxing which is what I'm most proficient in)

What would be easiest to observe in the physical world?

...

Definitely cars.

What data would I collect?

Date

Day of the Week

Time

Location

Weather

Make

Model

Car Type

Color

Approximate Year Range

Modified (Y/N)

Mod Type (None, Spoiler, Tint, etc)

Vehicle Class (Econ, Luxury, Performance)

(Well actually mod type and Modified would be redundant, cuz if it's not modified it'll just be None for mod type, but maybe it makes sense? Hmm... think about this later.)

**Do these attributes record variability?** I'd say yes. This is because we can capture a range of different Dates, Times, Locations, and Weather. Whether or not we find some kind of correlation between these variables and things like Car type, Color, Year range, etc, we'll find out in the test observation stage.

| Attribute               | Type                     | Description                    | Example                               |
|-------------------------|--------------------------|--------------------------------|---------------------------------------|
| Location                | Categorical              | Location observed              | Lombard<br>Walmart, Roselle<br>Metra. |
| Time                    | Temporal                 | Time observed                  | 6:24 PM                               |
| Day of the Week         | Categorical              | Day of Week of observation     | Monday, Tuesday                       |
| Date                    | Temporal                 | Date Observed                  | 10/4, 11/1                            |
| Weather                 | Categorical              | Weather when observed          | Partly Cloudy,<br>Sunny               |
| Make (brand)            | Categorical              | Make(brand) of car             | Toyota, BMW                           |
| Model                   | Categorical              | Model of car                   | Traverse, ES250                       |
| Car Type                | Categorical              | Type of car                    | SUV, Hatchback                        |
| Color                   | Categorical              | Color of car                   | Black, Red                            |
| Approximate Year Range  | (Ordinal)<br>Categorical | Estimated year interval of car | 2016-2020,<br>2018-2021               |
| Modified (Y/N)          | (Binary)<br>Categorical  | Aftermarket mods? (Visible)    | Yes, No                               |
| Mod Type                | Categorical              | Modification type              | Tint, Multiple                        |
| Vehicle Class (3 kinds) | Categorical              | Class of Vehicle               | Economy,<br>Luxury,<br>Performance    |

**Write a short description of what you want to observe and why you find it interesting. Then propose at least four initial domain**

**questions that you think your observations might help you investigate.** I want to observe cars and the environment they're in. I think there are many connections we could make between cars and, say, the state of the economy, local conditions, etc. Pertaining specifically to the data I'll be collecting in the table, my four domain questions would be:

- How does the distribution of vehicle types vary by location?
- Are older vehicles more likely to be modified than newer ones?
- Do certain brands have greater color variation between models?
- How does the distribution of Car Class vary by Car Make?

Of course if there isn't anything interesting, I can always come back and change these questions.

**Next, describe your proposed data collection process. Discuss:**

- **What constitutes one observation?**
  - One car recorded in the location's parking lot is one observation.
- **What attributes will you record for each observation?**
  - Listed in the table above
- **Where and when will you collect the data?**
  - Near/around my neighborhood, and downtown Chicago
- **Over how many locations, times, or days will you collect it?**
  - Over 4 days. Saturday, Sunday, Monday, Tuesday.  
Locations will be broken down into parking garages, Malls, Gyms, etc. This will be more up to my discretion and really is going to be so that I can make localized comparisons between locations I think would have actual variance. I intend to randomly selected between 10 locations that serve very different purposes.
- **How will you ensure that your data captures meaningful variation rather than a single snapshot?**
  - First, choosing different types of locations. Such as a gym, a local plaza, a store. Then the different times of day and days of week capture varying activity levels. Ideally this

captures car distributions across meaningfully different conditions. Also, when choosing data, I will assign each car a number between (1, number of cars), so that observations are easier to talk about.

- **How will you decide what to observe?**

- There's only so much that's visible when observing a car. Encompassing most of the physical attributes of the car is where the bulk of the valuable data lies, plus that's mostly what we can see anyway. The environment the cars are in will help us connect the car data to the environment and find correlations.

- **How will the collection be divided among group members?**

- Just me, since I'm doing this solo.

- **What might your collection process fail to capture?**

- It might fail to capture the distribution of cars at night, since most of my observations will be made in the day.

- **How might your collection process introduce bias?**

- The collection process will be done in the Chicagoland area, specifically Glendale Heights/Lombard. This means any influencing factors unique to the region (like income, car culture, etc) will be harder to spot, since there won't be observations from outside that region to compare to. Also since I'm choosing the places that *I'M* interested (since one person can't possibly hope to capture ALL the cars in their area via manual entry), some places, like grocery stores, might be more likely to have SUVs simply because they are more practical for storing groceries, or maybe Home Depot has more pickup trucks because the overlap between people going to home depot and people who have pickup trucks is noticeable.